

The Currency That Cannot Be Spent Alone: Co-Presence-Gated Redemption and the Chronicle↔Treasury Unification Circuit

The Mechanism of a Time-Currency Whose Redemption Is an Event Both Parties Must Attend

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Abstract

Time-banking and other time-currency systems have been practised since the 1980s, and they have tended to decay along a recognisable path: the hour becomes a unit of *service*, the service becomes an *errand*, and the system becomes a labor exchange with worse liquidity than money — at which point it dies of comparison with the thing it imitated. This paper locates the decay in one root: in the designs reviewed in §9, a time-credit is redeemed by one party *doing something for* the other, which makes the currency a wage denominated in awkwardness. The paper then specifies a design that removes that root. In the HeartBank® Chronicle, a pledged hour has exactly one redemption form: **an event both parties must attend**. Presence is defined as *time sacrificed for the other* — operationally, **synchronous shared attention**, in either of two geometries (*facing*: attention on each other; *flanking*: joint attention on a shared activity) and either of two attested modalities (in-person, proximity-attested; live call, session-attested). Asynchronous media cannot redeem an hour — a recorded message is the envelope that carries a pledge, never the event that spends it. The gate yields the currency's defining properties at a stroke: the redemption is symmetric in duration (both parties spend the same hour of finite time, which is why the mechanism can address loneliness at all: loneliness is not ended by receiving but by co-spending — while the quality and reciprocity of attention within the hour are neither required nor measured); absentee labor is structurally impossible (the gig-shape this paper identifies as the driver of time-banking's decay cannot be composed); and the delivered hour is attestable by both parties without trusting either alone.

Around the gate, the paper specifies the full circuit. The recipient authors the event and the giver holds a veto — the sacrifice being not merely of hours but of *agency over one's scarcest resource*. The curated event-menu spans togetherness and, under two guards (the **both-hands rule**: both parties perform; the **outward-benefit rule**: a third party benefits), side-by-side service — restoring shared work to the currency without readmitting extraction. Gratitude for a delivered hour crosses the wall into the money-currency, unprompted

and duration-blind, so the relationship never settles into barter; the shared event, uniquely witnessable because it is the one kindness with two people present, may be broadcast by its *beneficiary* — the camera inverted — and the community's answering money-gratitude routes by default to the time-giver. Unredeemed hours expire and genuinely die (the mechanic is the mortality thesis); their lapse occasions, never funds, a new anonymous gift. Nine claims (numbered 1–8, with 2a) are enumerated and dedicated to the public domain; seven predictions are pre-registered, including a standing tripwire (P-C3) that would detect the mechanism's own corruption — the emergence of a de-facto hour-to-money rate — and the authors commit to publishing it either way.

Keywords: time currency, time banking, co-presence, synchronous redemption, loneliness, dual-currency system, gift economy, redemption event, joint attention, proximity attestation, in-person key-fingerprint verification, mutual attestation, witness inversion, defensive publication.

Prior-Art and Non-Assertion Statement

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The following terms are coined in this paper and simultaneously freed with it: **co-presence-gated redemption**, **symmetric-cost redemption**, **facing/flanking** (the two geometries of shared attention), **envelope-not-delivery**, **the both-hands rule**, **the outward-benefit rule**, **sender-default routing**, **the tense wall**. Terms inherited from this corpus's earlier publications (the incommensurability-preserving coupler; witness-inversion; the two blindnesses; the spent-time floor) are cited, not re-claimed.

Time-currencies, service-credit systems, and mutual-credit networks constitute extensive prior art (§9) and are engaged generously; the present claims are confined to the event-layer architecture — the co-presence gate and its attestation, the authorship/veto composition, the guarded co-service class, the cross-currency circuit's routing, and the expiry mechanics. No systematic search of the world's prior art (a novelty census) has been run for this composition; the disclosure is published as prior art whether or not any element of it is new.

Division with sibling publications. *The Incommensurability-Preserving Coupler* discloses the wall-layer mechanisms (coupled but non-convertible dual currency, duration-blind recommendation, redemption-moment prompt suppression, moment-not-meter surfaces, ceremonial-unit anchoring, cross-currency response routing, the lapse channel); the position paper *What Money Can't Buy — and What Can't Buy Money* carries the institutional argument; the present paper discloses the event-layer mechanism: what a redemption is, who composes it, what gates it, how the circuit closes, and what the system refuses to build around it. The Chronicle is design-complete and unbuilt; every mechanism here is specified at the design layer.

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corrections; no claim added). Every version is preserved in the public repository history, and deposited versions are on Zenodo under the paper's concept DOI.

1 · The Problem: The Loneliest Loneliness, and Why Time-Banking Died of Labor

The epidemic is not what it looks like. The loneliness literature, following Weiss (1973), distinguishes *social* loneliness (the absence of a wider network of contacts) from *emotional* loneliness (the absence of a close attachment), and a wider network does not relieve the second. This paper applies the second category to the close bond that exists but has gone dark: the disconnected marriage, the estranged sibling, the diaspora child and the parent back home — relationships present in the address book and absent from the week. A mechanism for the first form matches strangers; a mechanism for the second must move *existing* bonds from dark to lit. The Chronicle targets the second form, and this choice does double duty: the counterparties are known (the trust-and-safety profile of reconnecting your own mother is categorically different from meeting a stranger), and the deepest need coincides with the lowest risk.

The adversary is the work-default. For most adults, time not deliberately assigned is absorbed by work — it pays the bills, and for some it is also the most tractable pleasure available. Presence must be *scheduled against* the default or it does not occur; sincerity is not the limiting reagent, calendaring is. A time-currency is, at bottom, a scheduling technology for love: the pledge converts a diffuse intention ("I should call more") into a dated, witnessed, expiring commitment.

And the graveyard is instructive. Edgar Cahn's time-dollars, the LETS networks, Ithaca HOURS, Japan's Fureai Kippu — the time-currency tradition is rich, humane, and largely sub-scale, and its recurring failure is *decay into labor exchange*: the credited hour becomes a claim on services (rides, repairs, errands), the system becomes a marketplace with money's obligations and none of money's liquidity, and participation collapses into the small population for whom awkward barter beats cash. The failure is architectural, not motivational. **Any time-credit redeemable as absentee service is a wage in embryo.** The design below removes the possibility at the type level: in the Chronicle, an hour cannot be redeemed *for* anything. It can only be redeemed *with* someone.

Connection to the unified mission frame. HeartBank's dual-currency thesis holds that money and time are the two scarcities of a human life, addressing the two great deficits — dignity (being seen) and loneliness (being with) — and that the two currencies are two tenses of one substance: money is stored lifespan, past tense, what one *has*; time is live lifespan, present tense, what one *is*. The Chronicle is the present-tense half. Its mechanism must therefore protect the one property money can never have — that the currency *is* the relationship, non-fungible, unspendable by proxy — and the co-presence gate of §4 is that protection made structural.

2 · The Canonical Scene

A daughter in Long Beach thanks her mother in Kâmpôt — for the recipe, for the phone calls she didn't return, for everything — and the thank carries a pledge of hours, its amount suggested by the system's

neutral agent, as her thanks always do. The pledges accumulate quietly below threshold; on her mother's birthday they cross it and activate: *four hours, redeemable for one month*.

Her mother redeems two of them: "*Walk the market with me Sunday morning. Bring your coffee.*" The daughter accepts — she could have declined this particular ask; she doesn't — and on Sunday morning in Kâmpôt — Saturday evening in Long Beach, fourteen hours behind in summer — they walk Psar Leu together on a live call, the phone riding in the mother's shirt pocket half the time, pointed at mangoes the rest. Nobody performs anything. It is not a message; it could not have been recorded in advance; it is two people spending the same hour on each other, and both ledgers mark it delivered because both were there.

That evening, unprompted — the application never asks — the daughter sends a small money-thank through the family's Treasury. It names the morning, not the meter: "*for the market, and the mango argument.*" Her mother, with her daughter's consent, shares a thirty-second clip of the walk; the aunts answer it, and their small re-thanks route to the daughter — the one who gave the hours. Two cousins, watching, pledge hours to their own mothers.

The other two hours are never redeemed. At month's end they die — the application does not soften this; the month was the point — and the system offers the daughter one thing only: the chance to give a small anonymous gift, funded from the commons pool, to a stranger nearby, in the memory of the hours that lapsed. The gift is new. The hours are gone. Both facts are the teaching.

3 • The Unit and the Wall

The unit. A Chronicle pledge denominates *time-gratitude*: hours promised as thanks, accumulated per-dyad, activated at a threshold, expiring one month after activation. The expiry is not an engagement mechanic; it is the product's thesis embedded in its unit economics — time is the currency that cannot be banked, and a promise of presence that can be deferred forever is not a promise. The unit's non-fungibility is absolute: only the pledger can deliver the hour, and only to the pledgee. There is no transfer, no exchange, no secondary market, no conversion. The currency is the relationship.

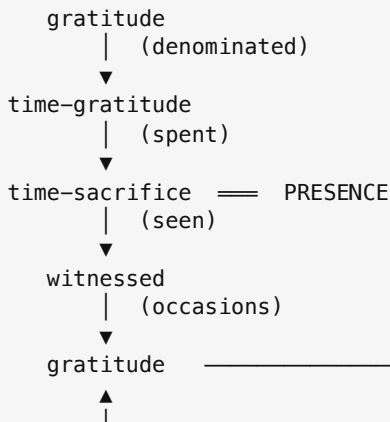
The wall. The Chronicle operates beside a money-currency (the Treasury), and the corpus's coupler doctrine governs the pair: coupled, never convertible. This paper adds the wall's deepest formulation, the **tense wall**: the two currencies are two tenses of one substance — life — and tenses do not convert. Stored past (money: crystallized, fungible, transferable) cannot purchase live presence (time: unfolding, non-fungible, attendable only in person, in the moment, by the person). One cannot pay someone to *have been there*. Every rate-prevention guard this corpus has published (no backward prompts at redemption; duration-blind recommendations; moment-not-meter language on every surface) is downstream enforcement of a wall that the metaphysics builds first.

Two published doctrines complete the unit's grammar and are inherited here: the **spent-time floor** — a delivered hour deserves gratitude as a fact, independent of any audit of the giver's motive, because spent lifespan is the one unfakeable cost — and the **two blindnesses** — the system stays blind to duration (no recommendation ever scales with hours) while the heart stays blind to motive (the thanker never withholds over purity doubts). Gratitude flows free of both meters.

3.1 • The loop the gate closes — the ground this paper shipped without

Added 2026-08-07. §§4–7 specified a gate and defended it mechanically; they did not answer the prior question a skeptical reader asks first — **why gate on co-presence at all**, rather than on anything else. This subsection supplies the ground. It introduces no mechanism, and the mechanism it grounds was already published; what follows is the argument that was in the design and not in the paper.

The mechanism sits inside a loop with six positions:



Gratitude, to move, must first be **denominated** — given a unit a person can actually spend. Denominated in time, it becomes a pledge. A pledge spent is **time sacrificed for another**, which is what presence is. Presence **witnessed** occasions gratitude in the other, and the loop turns again.

Mettā is the name of the loop and never a node in it. This is a design commitment, not a piece of vocabulary. Loving-kindness is what the whole circuit is *for* and what a person cultivates by running it — but the moment it becomes a position in the diagram, the ledger is being asked to record an interior state, and recording an interior state is a grading move. The architecture that refuses to price an hour cannot coherently score a heart. **Every node in the loop is an observable event; the disposition is the loop's name.**

Two consequences follow immediately, and both are already visible in the specification.

The Chronicle supplies exactly two of the six positions, because those are the two where the loop leaks. It **denominates** (gratitude that stays a feeling never moves; a pledged hour is gratitude given a spendable unit) and it **witnesses** (§6.4, §6.6). It does not supply presence, cannot supply the disposition, and deliberately does not supply the meeting.

Which is the strongest statement of what this system is not. It is not a calendar application, and the rule that keeps it from becoming one is *build the left side of the loop, never the right*: the Chronicle holds the pledge and the witness, and **never the appointment**. It does not know when you are meeting, does not remind you that you have not met, and cannot display an unkept intention — because a surface that renders an unkept intention has accused its user, and the accusation is exactly the outcome a gratitude system must

never produce. **The indictment surface is not moderated here; it is absent by construction, because the data that would populate it is never held.**

And the return leg is catalysed, not spontaneous. This is the loop's least obvious property and the reason the witness position exists at all. Presence that recurs at no cost stops occasioning gratitude — not because the recipient has stopped valuing it, but because nothing that arrives reliably and for free keeps interrupting the stream of attention. Steady love produces entitlement, and the entitlement is not a moral failure in the recipient; it is what habituation does. The corpus's own founding story illustrates it (one case, offered as illustration rather than evidence): the gratitude was immense and the giver never knew, because nothing in the situation *occasioned* its expression. **The witness is the catalyst that closes a loop which, unassisted, leaks at precisely that joint.**

Read from here, three mechanics specified earlier as design decisions are revealed as consequences rather than choices. **Expiry** (§7) is not a scarcity trick; it is the currency's physics — an hour is a commitment to spend a portion of a finite life, and a commitment that cannot lapse was never a sacrifice. **The tense wall** (§3) follows from the loop's direction: the circuit runs forward, and a mechanism that let value run backward would be paying for the past rather than funding the next turn. **The ceremonial unit** follows from the refusal to price: the loop's positions are events, and events are counted, never weighed.

Finally, this reframes the gate itself. Co-presence-gating is not merely a redemption rule. It is a **proof-of-sacrifice primitive** — an architecture for demonstrating that a finite, unrecoverable portion of a life was actually spent on another person — and it belongs beside this corpus's other two primitives, proof of humanity and proof of coordinate, as the third of a family. §7.1 states what that primitive is worth on a clock that is currently running, and the honest note is that **the primitive shipped before its rationale was written**: it was specified in July 2026 as a mechanism and named as a primitive only in August 2026.

(A relational-taxonomic route to the same conclusion — that no relation in the Abhidhamma's twenty-four joins two mental continua, so a system can supply occasions but never the arising — is developed independently in this corpus's Paṭṭhāna paper at §4.7. The arguments share no premises and reach the same design constraint, which is worth noting and not worth merging.)

3.2 · Why the unit takes no chain

A reader arriving from distributed systems asks a question this paper must answer before §4 can proceed. If the Chronicle issues a scarce unit, and if both parties must agree about what was promised and what was delivered, why is that unit not a token on a ledger of the usual kind?

Because the problem such a ledger exists to solve does not arise here. A blockchain solves **double-spending without a trusted third party**, and for double-spending to be a threat the asset must be transferable, storable, and forgeable by copying. The Chronicle's unit is none of the three. It is not transferable — only the pledger may deliver, and only to the pledgee. It is not storable — it expires one month after activation, and the expiry is the thesis rather than a parameter (§7). And it cannot be forged by copying, for the plainest reason available: **a person cannot be in two places at once**. The scarcity a proof-of-work chain manufactures at great expense, an hour of human presence already possesses.

	money	an hour of presence
transferable	yes	no – only the pledger, only to the pledgee
storable	yes	no – dies one month after activation
double-spendable	yes	no – one body, one place
∴ what it needs	SETTLEMENT (the present)	WITNESS (the past)

The objection worth taking seriously is that this proves too little. Time cannot be double-*spent*, but a promise of time can certainly be double-*made*: nothing prevents a pledger from committing the same Tuesday evening to three people, and that is a genuine failure. It is not, however, the failure a chain repairs. A ledger would record all three pledges with perfect fidelity and would not make Tuesday any longer. **Over-issuance is resolved at delivery rather than at issuance** — the co-presence gate of §4 is physically exclusive, so at most one of the three redemptions can occur and the other two lapse under §7 — and the constraint that resolves double-promising is the same physical constraint that made the chain unnecessary. One body, one place, one hour.

What the mechanism does require of a ledger is therefore not settlement but **witness**: that the pledge was made, that the meeting occurred, that the hour lapsed. Those are claims about the past, served by an append-only record whose roots are anchored for age — not by a consensus mechanism arbitrating rival claims to the same coin. Stated in one line: *money needs settlement, which is an action in the present; time needs witness, which is a claim about the past*. Which substrate carries which is an implementation fact of the wider architecture and not a feature of this mechanism.

4 • The Core Claim: Co-Presence-Gated Redemption

4.1 • The definition

A pledged hour has exactly one redemption form: an event at which both parties are present. Presence is defined as time sacrificed for the other — operationally, synchronous shared attention.

Each term is load-bearing:

- **Synchronous**: both parties in the same span of real time. Nothing prepared, recorded, or deferred can constitute the event.
- **Shared attention**, in either of two geometries:
 - **Facing** — attention on each other: the conversation, the meal, the walk where the walking is incidental.
 - **Flanking** — joint attention on a shared object or activity: cooking side by side, working side by side, watching the same fire. Joint attention is a bonding primitive as old as the parent and child looking at the same toy, and flanking is proposed here as the easier on-ramp for strained dyads (prediction P-C2 in §10 pre-registers the test) — a reconnection that cannot yet survive an hour of eye contact can survive an hour of weeding. Flanking is the doorway; facing is the destination; both redeem.

- **Two modalities, recorded but never ranked:** in-person (attestable by device proximity between two verified humans) and live audio/video call (attestable by the session itself). The remote modality is not a concession; the launch population is diasporic, and the mother in Kâmpôt is the design's first citizen.
- **Envelope, not delivery:** the recorded message — the B-LetterSM, the corpus's card-covered voice or video note — *carries* pledges and invitations; it never *constitutes* redemption. The distinction is structural: a system in which media can discharge a time-debt is a system in which presence has a recorded substitute, and the entire currency collapses into content.

4.2 • The symmetric cost

The gate has a consequence that none of the time-currency designs reviewed in §9 possesses: **redemption costs both parties the unit**. The recipient cannot receive the hour without spending an hour; presence is received only by matching it. Each party's attendance is a sacrifice for the other — the event is mutual by construction, and the asymmetry of giver and receiver survives only in the *claim* (who pledged, who authored) while dissolving in the *event* (both attend, both spend, both receive).

Current form. The symmetry the gate delivers is symmetry of *duration*, which synchronicity makes automatic; symmetry of *attention* is neither required nor measured, and an event in which one party gives attention the other cannot return is a fully valid redemption. §4.5 states the correction and claim 2 (§8) carries it. The paragraph above is retained as the originally disclosed variant.

This is why the mechanism can address loneliness at all, and the point deserves its plain statement: **loneliness is not ended by receiving. It is ended by co-spending**. A currency whose redemption were one-sided — a service performed, a message consumed — would deliver assistance to the lonely and leave the loneliness intact. The symmetric event is the anti-loneliness payload, and the gate is what makes it the *only* payload the currency can deliver.

4.3 • What the gate structurally excludes

The gig-shape — "redeem my hour: do this for me while I'm elsewhere" — cannot be composed in this system. There is no redemption type in which one party labors and the other is absent; absentee service is not discouraged but *inexpressible*. This single exclusion severs the decay path of §1: the currency cannot become a labor exchange because labor-for-an-absent-beneficiary is outside its grammar. (Labor *alongside* a present partner, for a third party, returns under guard in §5.3 — the distinction between working *for* someone and working *with* them is exactly the distinction the gate enforces.)

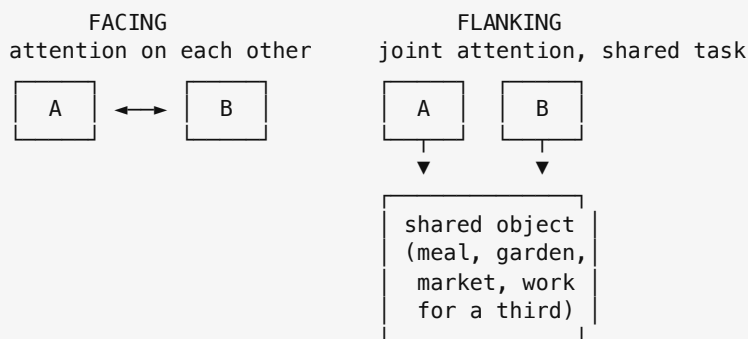
4.4 • Attestation

Delivery is marked by both-party confirmation, and the co-presence gate makes the confirmation *attestable*: an in-person event is corroborated by device proximity between two verified humans (the corpus's existing proximity primitive and Proof of HumanitySM layer); a remote event is corroborated by the live session both parties held. Neither party can unilaterally fabricate a delivered hour; a dyad colluding to fabricate one harvests — as §11 details — nothing the system pays for. The **dual public ledger** completes the accountability grammar: hours *given* (delivered) and hours *received* stand as two public axes — credibility

and belovedness — while chronic non-delivery and chronic decline price themselves reputationally without any tribunal.

Current form. The ledger is no longer specified as an enforcement surface. As now specified, the dyadic content of the record is private and only proofs are public; the gap between hours received and hours delivered is never computed for display, and chronic decline or non-delivery is never rendered. Honoring is enforced by expiry and by the giver's free refusal (§7.2), and claim 3 (§8) carries the withdrawal. The sentence above is retained as a disclosed variant.

THE REDEMPTION EVENT – the only spendable form



modalities: in-person (proximity-attested) · live call (session-attested)
excluded: anything asynchronous (envelope, never delivery)
cost: symmetric – BOTH spend the hour; that is the point
attested: both-party signoff; neither party trusted alone

4.5 · The asymmetric case, and a correction to claim 2

Added 2026-08-07, and it tightens a published claim rather than extending one. The correction is stated first because the prior wording overstated the mechanism.

The question that forced it: does the receiving party sense the gift when they cannot return it — the declining parent, the depressed friend, the small child, the estranged sibling who agreed to come but not to reciprocate?

The answer is yes, and it splits an operation this paper had treated as one.

B senses the attention. B infers the sacrifice.

These are different epistemic acts. **Attention lands directly** — it arrives through the senses at full bandwidth, and a person can tell, without reasoning about it, whether they are being attended to. **Sacrifice cannot land directly, because the foregone alternative is never in the room.** What A gave up by being here is by definition absent; it is reconstructed from signs — the distance travelled, the workday moved, the phone face-down. Sacrifice is a **counterfactual**, and counterfactuals are inferred.

Symmetry is therefore not required for the mechanism to work. If influence between two people runs as two independent one-way edges rather than as a single bond, then A's attention reaching B is *one edge*

functioning exactly as specified, whether or not the return edge carries anything. **Asymmetric presence is not a degenerate case of the mechanism; it is the mechanism running on one side.**

Which means claim 2, as published, was too strong. The claim read "*presence received only by matching presence*," and §4.2 spoke of both parties dissolving into the event. **The gate does not, and cannot, deliver that.** What the gate actually enforces is that both parties spend the same *duration* — that much is automatic, since the event is synchronous and each party's hour is gone either way. What it cannot enforce, and must never attempt to, is that both parties spend the same *attention*. Claim 2 is accordingly restated in §8 as a claim about **symmetric duration with asymmetric attention permitted**, which is weaker in wording and considerably stronger in truth. **The prior-art clock for the restated claim 2 runs from the 2026-08-07 push.**

"Cannot return" is three cases and they want different treatment.

Case	What is true	Design response
Cannot match, but can acknowledge	the return edge runs; it simply does not carry an hour — and receiving well is itself a giving	the majority case: serve it. No surface may treat acknowledgment as a lesser return
Will not return	the estranged party who attends and gives nothing back	the case the mechanism was built for; nothing here is conditional on their reciprocating
Cannot perceive	the unconscious, the very ill, the dying	the gift is complete anyway — a position this corpus already holds from both directions, in the unsent letter and in the letter whose reader never knew

The design consequence is the section's real payload, and it inverts how the asymmetry should be read. The gate makes symmetric duration automatic and symmetric attention **impossible to require**. A mechanism that tried to require it would have to measure it, and measuring the quality of someone's presence is the one thing this architecture must not build. **So the asymmetry is not a failure mode to be closed. It is the space where the gift lives** — because a gift that could be compelled from the other side would not be a gift, and the whole design rests on that distinction.

And one honest limit hardens into a refusal. Because sensing is inference, it fails in both directions: **performed attention reads as real, and flat affect reads as absence.** A system that scored presence would therefore reward performance and penalise atypical expression — penalising, most reliably, the autistic, the depressed, the grieving, and the very people the product exists to reach. **No surface of this system may score, rate, or infer the quality of anyone's presence.** This is stated as a refusal rather than a limitation, and it is recorded in the corpus's directive backlog as a hard constraint on the autonomous agent, so that no future optimisation can discover it as an available signal.

(What the mechanism supplies is conditions, never outcomes: one may stage the occasion and cannot deliver anyone a good hour. §3.1's loop states the same boundary from the other end.)

The third case has a path, and this paper already contained it. *(Added 2026-08-26; it resolves an open edge rather than extending a claim, and no numbered claim changes.)*

The row above says the gift is complete anyway and stops there. That is true, and it leaves the *mechanism* with nothing to do — which has read, since publication, as the gate having no answer for the case the corpus calls the purest gift. It has one.

The move is to stop treating the beneficiary as a party to the event. The one who cannot attend is not a party to the co-presence at all. **Two others form the co-present dyad** — two siblings at a bedside, two children at a grave — **and the one who cannot attend is the third-party beneficiary.** This is the co-service class of §5.3 running unchanged, and it passes both of that section's guards without amendment: **both-hands** (both attend, both sit, symmetric effort — no asymmetry through which anything could be extracted) and **outward-benefit** (the beneficiary is a third party, never the redeeming dyad's own advantage).

It is not substitution, which this mechanism refuses elsewhere and refuses here. §7's treatment of a lapsed hour is explicit that a stranger's hour is not the drifted brother's hour; nothing in this path swaps one recipient for another. **The flanking pair supplies the mutuality the beneficiary cannot supply, and the gift's address does not move.** §5.3's outward-benefit rule already seats the beneficiary *outside* the co-present dyad — that is what the rule is for, and this is the shape it was written to permit.

Two consequences follow with no further machinery. Directed at a parent or an elder, the same act carries **apacāyana** (reverence) atop **veyyāvacca** (service), so the merit-structure deepens rather than strains. And **the dead are reached by the identical structure** — two people at a grave together — which makes the ancestral-observance traditions, Pchum Ben among them, **native instances of this mechanism rather than adjacent practices** it has to be reconciled with.

What this does not close, stated plainly because the limit is where this paper's own citations live: the solitary case. The path requires two attendees. It has nothing for the only child alone at the bedside, or for the person who learns of a death a year later — **and both of the cases cited above for the gift is complete anyway are solitary:** the unsent letter, and the letter whose reader never knew. The flanking path widens the *accompanied* case and leaves the solitary one exactly where it was. **It must not be closed by relaxing the gate;** a proxy attendee remains absentee-shape, which is the precise failure the gate exists to prevent.

And the path creates one guard, which is stated here so it is not discovered later as an available feature. A beneficiary who cannot attend must **never become a destination in the curated menu of §5.2.** Memorial and sacred sites are contemplative-only across this architecture, and a menu offering "*visit someone who cannot respond*" would render a person as **an occasion for someone else's redemption** — the field-into-lack inversion this system refuses on every other surface. **The visit is something people bring to the menu; it is never something the menu proposes.**

4.6 • Proof of Meeting, and the pretext problem

Added 2026-08-22. This section adds no claim. It supplies a mechanism §4.4 named without specifying — offline corroboration that two people were actually together — and then makes an argument about why that mechanism, which is thirty years old and free, is so rarely performed by ordinary users. The mechanism is prior art and is cited as such. The argument is the only thing here that is ours.

The gap. §4.4 states that an in-person redemption is "corroborated by device proximity between two verified humans," which is true and underspecified. Proximity attested *how*? Two answers are ordinarily

available, and both are weak in the same place. A **coordinate** — GPS, a geofence, a check-in — is a gate rather than a proof: it is spoofable with a free app, and this corpus has already ruled that geolocation may gate a thing and may never prove one. A **server-mediated check-in**, where both clients report to the same backend within a window, is stronger against casual forgery and puts the ledger's most intimate event at the mercy of connectivity in the exact moment two people have gone somewhere to be undisturbed.

A third answer exists and is old. Proof of Meeting SM (PoM) is an in-person exchange of public-key fingerprints between two devices — free, local, mutual, symmetric, non-transitive, requiring no account, no coordinate, and no network. Two phones in short-range contact each end holding a record, signed by the other, that this pair of devices was co-located at a moment. The property that makes it interesting for this mechanism is negative and blunt: **a GPS faker cannot forge a second human's handset**. Presence is attested by another party's device rather than by a sensor the attesting party controls.

Four things it is not:

PoM is not	Because
identity	a fingerprint binds a <i>device</i> , and devices are lent, sold, and stolen
location	it attests <i>proximity between two devices</i> , and names no place on earth
humanity	that is a separate primitive (Proof of Humanity SM) and is not implied by contact
safety	a meeting is not a character reference. Nothing whatsoever is asserted about the other party

The prior art is decisive, and it is the whole point of this section. PoM is the **key-signing party**. In-person verification of public-key fingerprints is the founding social ritual of PGP's web of trust (Zimmermann, 1991) — documented in V. Alex Brennen's *Keysigning Party HOWTO*, which also describes the Zimmermann–Sassaman method for verifying fingerprints efficiently in large groups, and practised at conferences for three decades. Its living descendants are ordinary infrastructure: **Signal's safety numbers**, compared or scanned in person to confirm a conversation's keys; **SSH host-key pinning** and the trust-on-first-use model that carries most of the world's server administration. We claim none of this. PoM is a **naming and a scope rule** over a mechanism the security community built, published, and freed long before this institution existed.

Which reframes the question entirely. It is not whether in-person key verification works. It is why almost nobody does it.

The usability literature answers that, and it does not answer *cryptography*. Vaziripour and colleagues put 36 pairs of participants in a room with the authentication ceremonies of WhatsApp, Viber, and Facebook Messenger and found that, when instructed only about potential threats, **participants located and completed the ceremony in 14% of attempts**; when also told of the ceremony's importance, **79% did**, taking on average 3.2 minutes to find it and a further 7.8 minutes to complete it (*Is that you, Alice?*, SOUPS 2017). A follow-up modified Signal to prompt users toward the ceremony and simplified the ceremony itself; with no training, **90% of participants using the prompted version completed it, at a median of two minutes** (*Action Needed!*, SOUPS 2018).

Read those two results side by side and the diagnosis is not competence — it is occasion. People perform the ceremony in two minutes when something in front of them supplies a reason to; left to themselves they neither find it nor look for it, because the ceremony is not part of anything they came to do.

That is the pretext problem, and this mechanism happens to solve it. A key-signing party is an event whose entire purpose is the verification — the pretext is the ceremony, which is why it has remained, for thirty years, an activity for people who enjoy cryptography. **Key verification never failed on the mathematics. It failed on the invitation.** What the mechanism specified in this paper supplies is a reason to be in the same room that has nothing to do with keys: an hour that somebody pledged, redeemed the only way it can be redeemed — in person, both parties spending the same finite unit. The verification then rides an occasion the participants already wanted, and costs a gesture inside a meeting that was going to happen anyway.

Stated generally, because it is not about this product: **a verification ritual with no host behavior is a ritual nobody performs, and the repair is not a better protocol but a better pretext.** Thirty years of excellent protocols and no pretext is the experiment that has already been run.

The fit is exact, and for a reason worth stating rather than assuming: PoM proves the *meeting*, never the *content* of it. That is ordinarily read as PoM's weakness. Here it is the requirement. Co-presence gating needs *they were together*; it must never learn *how it went*, because §4.5 has already made the quality of anyone's presence unmeasurable by refusal. A proof that could report on the hour would hand this system precisely the signal it has committed never to hold.

The governing rule, and it is a prohibition. PoM is good for access and bad for value. Its forgery cost is *one real meeting* — and two colluders can pay that cost daily, over coffee, indefinitely. So it may be used to decide whether a pair may be admitted, whether a redemption may be marked delivered between these two parties, whether a roster or a circle is open to someone; and it **must never mint, price, score, or rank**. This is the play/currency wall in a second domain: only witnessed giving moves the currency, and a proof of contact is not a witness to a gift.

What it does not replace. Both-party signoff (§4.4) stands unchanged: PoM corroborates the *where*, the signoff carries the *whether*, and neither party is trusted alone in either. Nor does anything here weaken §11 — a dyad fabricating a delivered hour still harvests nothing the system pays for, which is exactly why a proof with a one-meeting forgery cost is tolerable at this joint and would be intolerable at a joint that paid out.

Current form. Proof of Meeting is offered here as a candidate corroboration mechanism, not an adopted one: the institution has ruled on the primitive and its scope — it may decide access, never value, and it may never mint, count or rank — but not on its use as this gate. And a fabricated hour is not strictly costless: a delivered hour registers as one crossing in the aura, and the aura weights a bounded share of the commons pool's annual disbursement, so it is the bound on that share, not an absence of value, that keeps a one-meeting forgery cost tolerable here. The text above is retained as disclosed.

Honest limits. The primitive is unbuilt and $n = 0$ here. It attests *contact*, not *duration*: a proof exchanged in eight seconds and a proof exchanged inside a two-hour walk are the same artifact, so the gate continues to

rest on the event and the mutual signoff rather than on the proof. It presumes two working handsets with functioning short-range radios, which is an accessibility floor and not a small one — the design must keep a non-device path to marking delivery, or it will have excluded the poorest and the oldest participants from the only spendable form of the currency. And the pretext argument above is a **hypothesis about behavior**, not a result; it is stated so it can be checked.

5 • Composing the Event

5.1 • The recipient authors; the giver may refuse

The recipient of pledged time chooses how it is spent — the inversion of ordinary gift-giving, where the giver picks. The doctrine beneath the design choice: the pledge's sacrifice is not merely of hours but of **agency over one's scarcest resource**. To give an hour *and the say over how it is spent* is the difference between donating time and surrendering it; the inversion is where the gift's weight lives. The giver retains a per-request veto — consent is mutual at every event, and a refused request costs the refuser only what the public ledger honestly shows about chronic refusal.

Current form. No count of refusals, chronic or otherwise, is displayed (§7.2; claim 3 as amended). A refused request costs the refuser nothing on any surface; the veto is free, and repeated declining is answered by expiry and by the mutual veto itself rather than by a displayed record. The sentence above is retained as a disclosed variant.

5.2 • The curated menu

Event composition draws from a curated registry of *togetherness* — shared meals, walks, calls, visits, ceremonies, the co-service class below — and not from a free-text task box. The curation is not paternalism; it is the grammar made visible: the menu answers "what can an hour be?" with forms of *with*, never forms of *for*.

5.3 • The co-service class: shared work, under two guards

The menu includes work — chores and volunteering — under exactly two conditions, jointly sufficient to keep the labor-bar intact:

1. **The both-hands rule.** Both parties perform the work, side by side, symmetric in effort. Extraction requires asymmetry — one directs while the other labors; when both hands are in the soil, it is service, and it redeems.
2. **The outward-benefit rule.** The work's beneficiary is a third party — the temple grounds, the elderly neighbor, the riverbank — never the redeeming dyad's own advantage.

Under both guards, even "paint my mother's fence together" composes cleanly: shared filial service, flanking-mode, outward-directed — and for the estranged-family dyads at the Chronicle's core, quite possibly the best event in the registry. The class also travels the remote modality (two people transcribing manuscript pages together on a call are flanking across an ocean). What the guards exclude is precisely and only the laundered errand: work directed by a present-but-idle beneficiary, or work whose benefit circles home.

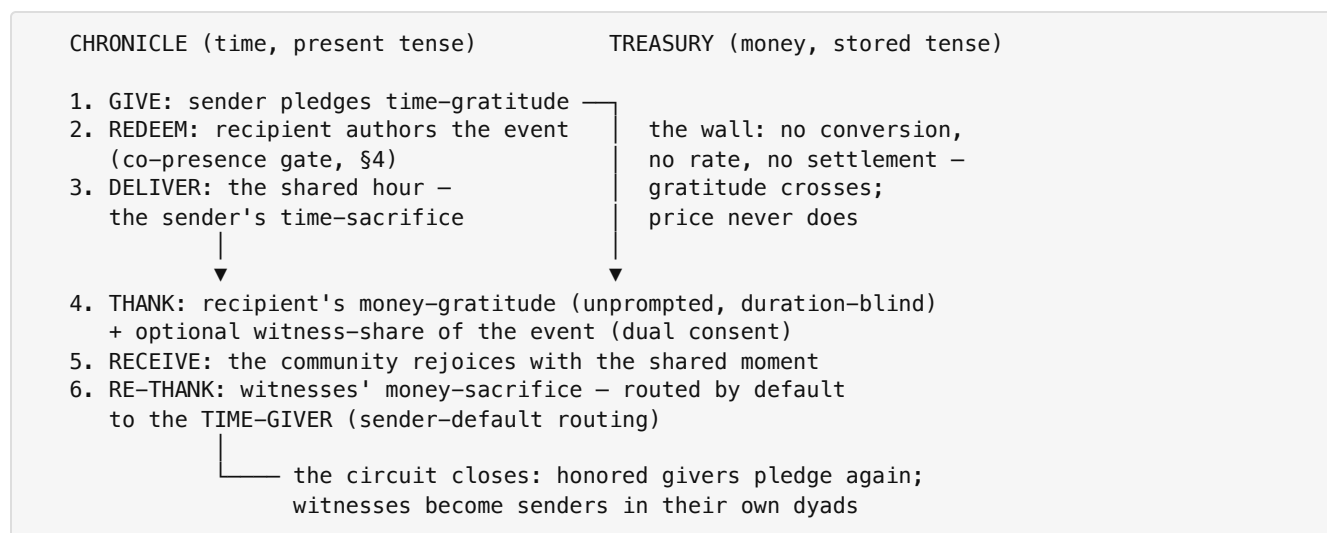
5.4 • The safety set

The known-counterparty core (reconnection, not stranger-matching) does most of the safety work by scope; the remainder is architectural: mutual veto on every event; the curated menu's structural exclusion of extraction; coercion-safe presence signaling (availability expressed as a curated menu, mutuality as a signal never a gate, block-and-override always); and the anonymity boundary — the Chronicle's stranger-facing layer is a separate, separately-published mechanism with its own safeguards, walled off from the reconnection core by an explicit flag, and nothing in this paper's mechanism ever routes a stranger to a doorstep.

6 • The Circuit: How Gratitude Crosses the Wall

6.1 • The six steps

The Chronicle and Treasury close into a single circulation — the unification circuit — whose canonical form is:



6.2 • The joints are canonical

Each arrow in the circuit is an instance of gratitude-that-acts — the classical pairing of *knowing* what was done with *making it known* in deed (the kataññū-katavedī structure, AN 2.31–32, engaged here as grounding). The same hour changes face across the circuit: at pledge it is gratitude (backward-looking recognition); at delivery it is sacrifice (forward-looking gift). One object, two faces, separated by time — the recognition-then-enactment joint rendered as a state machine.

6.3 • Why the response crosses currencies

The thank for a delivered hour lands in *money*, and the reason is the wall itself: same-currency response (time-for-time) would make the relationship a scheduling ledger — barter with extra steps, each hour begetting an owed hour forever — and would re-consume the giver's protected scarcity as the price of having given it. Cross-currency gratitude can never settle, because no shared unit exists in which the

account could be squared; the unpayable response is what keeps the relationship *open*, which the gift literature, from Mauss onward, has identified as the gift's actual function. The routing is inherited from the coupler publication's claims; what this paper adds is the circuit position: the cross-currency response is step 4 of a loop, not a bilateral courtesy.

Three constitutional guards, previously published, hold the crossing uncorrupted and are restated as inherited: the application **never prompts** the money-thank at or after redemption (the flywheel is a description of what love does, never a funnel the interface drives); the neutral agent's money-thank recommendation is **duration-blind**, never arithmetic on hours; and gratitude language on every surface names **the moment, not the meter** — "the morning at the market," never "the two hours."

6.4 • The witness inversion, and sender-default routing

Most kindness is done alone, which is why self-recorded kindness dominates gratitude ledgers and why performative capture is its permanent risk. The co-present event is the structural exception: **it is the one kindness with two people present, so the beneficiary holds the camera**. Content about the event is captured *by the grateful party, about the giver* — testimony, not performance; the recording is itself an act of gratitude. Its broadcast is disciplined (dual consent always; a curated, rationed gallery, never an open feed; the success metric is reconnections-inspired, never watch-time) and its response flow is claimed here as mechanism: **community re-thanks route by default to the time-giver** — the witnessed deed's author — while the witness's own act of capture earns whatever gratitude it organically draws. The default is a routing rule, not a split: no forced division, no secondary market in witness credit.

6.5 • What closes the loop

Nothing in the circuit mints. The giver ends the loop with delivered-hours honor, received money-gratitude, and an intact relationship; the witness community ends it having *spent* (money-sacrifice, step 6); and the loop's growth engine is the scene's quiet last line — the cousins who watched a reconnection and pledged hours to their own mothers. The circuit recruits by witness, not by referral bonus; what it compounds is dispositions.

7 • Threshold, Expiry, and the Death of an Hour

Threshold-activation. Small thanks accumulate as pledged time without expiry pressure; at a threshold the accumulated pledge activates into a redeemable, expiring commitment. The two-stage design keeps casual gratitude light (no one owes a coffee date for a small thanks) while making accumulated gratitude *real* (at some point, the thanks want a Sunday).

One-month expiry, and the hour genuinely dies. An activated hour unredeemed after one month is lost — not banked, not transmuted, not softened. The individual hour has no afterlife: Chronicle time is dyadic and non-stored, an expired commitment is not a deposit, and "your hour flowed onward" would be a lie the system refuses to tell. The grief of the lapsed hour is the product's teaching — *redeem before it is too late* — and the mechanic is the message.

The two-layer resolution, and the lapse-channel. At the aggregate layer, the platform's revenue funds the commons pool as a platform-level flow scaled to system-wide activity — never a per-hour transmutation, never a user charge. At the per-event layer, a lapse *occasions* (never funds-from-the-hour) an offer to the giver: to direct a small, anonymous, pool-funded money-gift to a nearby verified stranger, in the memory of the hour that died. The gift is new; the amount is agent-set and duration-blind; the delivered event remains the visibly superior outcome. The channel — previously claimed in the coupler publication as the lapse-channel — completes the pledge's incentive shape: the worst case of pledging time is becoming the human hand of an anonymous gift, which makes the pledge downside-free without making the lapse rewarding.

7.1 • The one unit that cannot be delegated

Added 2026-08-07. This subsection states what the gate is worth against a substitution pressure that is currently live, and it is the paper's most time-sensitive argument.

Most other components of companionship are now substitutable in some measure, and the substitutes are improving quickly. Conversation, attentiveness, memory of what you said last week, patience without limit, availability at three in the morning — a machine can supply all of it, in many cases better than a tired human can, and the companionship-application market is being built on exactly that observation.

Finite duration is the one component that cannot be delegated, and the reason is structural rather than sentimental.

Presence-as-experienced can be counterfeited. Presence-as-sacrifice cannot.

The experienced surface of presence — warmth, attention, responsiveness — is a set of observable behaviours, and observable behaviours can be produced by whatever produces them best. But the *sacrifice* underneath is not a behaviour. It is the fact that a being with a finite and unrecoverable allotment of hours spent one of them here, and therefore did not spend it anywhere else. **The value of the hour is the foregone alternative**, and a system that can run any number of parallel instances has no foregone alternative to give up. It is not that machine time is cheap; it is that **parallelisability removes the scarcity structurally**, so there is nothing for a sacrifice to consist of.

This is why the gate is a **proof-of-sacrifice primitive** and not merely a redemption rule. It is an architecture for demonstrating that finite life was actually spent — which is precisely the property that survives when everything experiential about presence becomes reproducible.

Two boundaries keep the argument honest.

The claim governs what may denominate the currency, never what may help people. A machine companion that reduces someone's suffering at four in the morning is doing something good, and nothing here says otherwise. What it cannot do is *mint an hour in this ledger*, because the ledger's unit is defined as spent finite life. Confusing the two would turn a claim about accounting into a claim about worth, and this paper makes no claim about worth.

And the argument is falsifiable, which is unusual for an argument of this shape. It rests on three properties of present-day systems — parallelisability, forkability, and non-mortality. **An agent that was**

single-threaded, unforkable, and mortal would break the premises, and the honest position is that such an agent would have a genuine claim to the unit. The argument is contingent on architecture, not on substrate, and it is stated so that it can be checked rather than assumed.

7.2 • Enforcement is the clock, not the audience

§7 establishes that an activated hour dies unredeemed after a month. That fact does a second job this paper has not previously claimed for it, and the claim withdraws a weaker mechanism the corpus had been carrying elsewhere.

The dual public ledger of time-given and time-received was specified to enforce honoring by visibility: a participant who chronically receives more than they deliver becomes legible as such, and is offered fewer hours thereafter. The mechanism would probably work. It is nevertheless the wrong one on this corpus's own commitments, because it enforces by **rendering an absence** — and a rendered absence is an accusation. Such a surface makes the viewer's judgment the engine, requires an audience to be present at the moment the guard is tested, and asks a participant to feel a debt in the one currency the architecture exists to keep out of debt's grammar.

Expiry requires none of that. An unredeemed hour is simply gone: the pledger's capacity to make good was consumed by the clock rather than withdrawn by a crowd, and the giver's standing refusal (§5.1) already prices repeated declining without anyone being displayed as delinquent. **The enforcement is a property of the unit rather than a rule about display, and it holds with nobody watching**, which is the test this corpus applies to its guards. What a ledger may show is what a participant has given and what they have received. What it may never show is the gap between them.

The cost is real and is not concealed. Visibility would deter a participant who chronically receives and does not deliver faster than expiry does, since expiry reaches them only through the quiet accumulation of hours that no one offers. The slower mechanism is accepted deliberately: the faster one is a shaming instrument, and this is a product about being seen generously.

Current form. Where this paper says a ledger may show what a participant has given and received, the audience is now specified. The record is one view of an append-only log whose rows are private and whose proofs are public; no per-participant total of hours given or received is published, and the public cross-participant signal is the aura, which records each delivered hour as one crossing, never as a quantity. A participant may see their own history, exported complete and signed; nothing in it is ranked, compared with anyone else's, or counted against them, and the difference between hours received and delivered is never computed for display, to them or to anyone. Public per-participant totals are retained as a disclosed variant.

8 • The Claims

Enumerated as prior art; each claimed severally and in combination:

1. **Co-presence-gated redemption:** a time-currency in which the sole redemption form is a synchronous shared-attention event attended by both pledger and pledgee — in either geometry (facing or flanking)

and either attested modality (in-person via device-proximity between verified humans; remote via live session) — with asynchronous media structurally excluded from redemption (envelope-not-delivery).

2. **Symmetric-duration redemption with asymmetric attention permitted:** the property, and its deliberate exploitation, that the redemption event costs both parties the same **unit of finite time** — enforced automatically by synchronicity — while the **quality and reciprocity of attention within the event are neither required, measured, nor scored**, so that an event in which one party gives attention the other cannot return is a fully valid redemption rather than a degenerate one. (*Restated 2026-08-07; the original wording — "presence received only by matching presence" — asserted a symmetry the gate does not deliver and must not attempt to enforce. See §4.5. The prior-art clock for this restatement runs from the 2026-08-07 push; the claim as originally published is superseded, not withdrawn.*) 2a. **The presence-scoring refusal:** the design commitment that no surface of the system scores, rates, ranks, or infers the quality of any participant's presence or attention — grounded in the finding that presence is sensed but sacrifice is *inferred*, so that any such measure necessarily rewards performed attention and penalises atypical affect.
 3. **Recipient-authored, giver-vetoed event composition:** the redeeming party authors the event from a curated togetherness registry; the pledging party holds a per-event veto, with both-party delivery signoff. **Amended 2026-09-02** — this claim previously continued "*chronic refusal and chronic non-delivery are priced reputationally by a dual public ledger (given/delivered $\times$ received)*". **That half is withdrawn** (§7.2): the gap between received and delivered is never computed for display, and enforcement is carried by expiry and by the giver's free refusal rather than by rendering a deficit. What survives is the composition mechanism itself; a ledger may show what a participant has given and received, never the difference.
 4. **The guarded co-service class:** shared work as a redemption event under the conjunction of the both-hands rule (symmetric performance by both parties) and the outward-benefit rule (third-party beneficiary), composable in-person or remote — restoring service to a time-currency while structurally excluding labor extraction.
 5. **The cross-currency response circuit:** a dual-currency architecture in which gratitude for a delivered time-event is expressed in the money-currency (unprompted, duration-blind, moment-not-meter — wall-layer guards inherited from the coupler publication) as one step of a closed six-step circulation spanning both currencies.
 6. **Witness-inversion broadcast with sender-default routing:** event-witness content captured by the beneficiary about the giver, shared only under dual consent into a curated rationed gallery measured by reconnections-inspired rather than watch-time, with community re-thanks routing by default to the time-giver.
 7. **Threshold-activation with mortal expiry and the two-layer lapse resolution:** pledge accumulation below an activation threshold; one-month expiry after activation with the individual hour irrecoverably extinguished; aggregate platform-revenue funding of the commons pool decoupled from any individual hour; and lapse-occasioned giver-directed anonymous giving (lapse-channel per the coupler publication).
 8. **The composition:** claims 1–7 as a single system — a time-currency that cannot be spent alone, cannot become labor, cannot convert to money, and closes its circulation through witnessed gratitude.
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9 · Prior Art, Generously

The time-currency tradition. Edgar Cahn's time-dollars and the TimeBanking movement (the founding articulation of time as egalitarian currency, and the co-production ethic this design honors); LETS mutual-credit networks; Ithaca HOURS; Japan's Fureai Kippu caregiving credits (often cited as the tradition's most successful instance, and the closest ancestor in spirit — care delivered to elders, though redeemable as absentee service and transferable across parties, both of which this design excludes); Bernard Lietaer's complementary-currency scholarship; the WIR cooperative. The failure analysis of §1 is offered with respect: these systems discovered the demand; this design responds to their decay mode.

The witnessed-conversation precedent. StoryCorps — two decades of recorded conversations between people who love each other, publicly broadcast — proves mass demand to witness real reconnection. The distinction: StoryCorps archives the conversation as its terminus; the Chronicle's witness layer is one step of a circuit whose metric is the *next* reconnection it inspires, and whose recorded artifact is never the redemption itself.

Synchronicity and presence mechanics. BeReal's synchronous-authenticity prompt (the interaction pattern of unfakeable simultaneity); video-presence verification in dating and identity platforms; BLE proximity attestation (the contact-tracing engineering wave); shared-activity platforms (watch-together, exercise-together features) as flanking-mode precedents in commercial form.

In-person key verification. PGP's web of trust and the key-signing party ritual it produced (Zimmermann, 1991; V. Alex Brennen's *Keysigning Party HOWTO* and the Zimmermann–Sassaman efficient group method); Signal's safety numbers; SSH host-key pinning and trust-on-first-use. §4.6 takes the mechanism from this lineage wholesale and claims nothing in it; what §4.6 adds is the usability finding that the ritual fails on **occasion rather than on competence** (Vaziripour et al., SOUPS 2017 and 2018) and the observation that a redeemed hour supplies a pretext for meeting that is not about keys, which the key-signing party, whose pretext is the verification itself, does not.

The gift and its literature. Mauss on the obligation-structure of the gift; Hyde on the gift that must move; the sociology of earmarked monies (Zelizer); the crowding-out literature (Titmuss; Gneezy & Rustichini) that grounds the wall's necessity. Buber's I–Thou stands behind the facing geometry; the joint-attention literature of developmental psychology behind flanking.

This corpus. The incommensurability-preserving coupler (wall-layer claims; Terra/Luna as the coupled-with-convertibility anti-example); the gift-tag-time-reveal publication (the pledge's physical on-ramp); *The Grace That Settles Nothing* (the coupler's third instance and this paper's sibling in the mints-nothing family); the Heart volume's published circuit doctrine and pre-registered P3–P7; the recommendation-function methodology publication (the neutral agent's band-clamp seam).

The clock. Patent activity around agentic payments — automated agents negotiating, scheduling, and settling on humans' behalf — is, in the authors' assessment, the live enclosure risk for exactly this territory: an agent-scheduled "quality time" feature with dynamic pricing is this mechanism with the wall removed, and the reason this publication is timed to precede the wave's arrival at the household.

10 · Pre-Registered Predictions

Stated 2026-07-21, before the Chronicle exists; P3–P7 of the Heart volume (initiation/response directionality, anchoring, age-shape) remain in force and are not restated.

- **P-C1 (the event outlasts the exchange).** Dyads completing at least one co-present redemption will show greater reconnection persistence at ninety days (continued contact beyond the platform's mechanics, self-reported "understand each other better") than dyads with pledges and messages but no redeemed event. Falsified if pledge-only dyads reconnect at equal rates — which would imply the gate is ceremony, not mechanism.
- **P-C2 (flanking is the on-ramp).** Among long-estranged dyads, first redemptions will over-select the flanking geometry relative to facing, and the flanking share will decline within dyads across successive redemptions. Falsified if first events show no geometry preference.
- **P-C3 (the standing tripwire — the wall holds socially).** Across the population of post-redemption money-thanks, thank *amounts* will show no significant correlation with event *duration*. This is the mechanism auditing its own deepest guard: a positive correlation is the empirical signature of a de-facto hour-price forming despite the wall, and the authors commit to publishing the correlation, whichever way it runs, at every reporting interval. The wall is not proven by architecture; it is proven by this number staying null.

The loop predictions (added 2026-08-07, before the Chronicle exists). §3.1 claims the loop leaks at two joints and that this mechanism is a catalyst at both. That is testable, and the second of the three is the design's founding bet.

- **P-L1 (denomination catalyses).** Dyads who convert an expressed intention into a *denominated pledge* will reach a completed co-present event at a higher rate than dyads expressing the same intention undenominated ("we should catch up soon"), measured over ninety days. **Falsified** if conversion rates are equal — which would mean the unit is bookkeeping rather than a catalyst, and the pledge layer is decoration on a calendar.
- **P-L2 (witness catalyses the return leg — the founding bet).** Among completed redemptions, **witnessed** events will produce more downstream gratitude expression and more subsequent giving than unwitnessed events of equal duration and comparable dyad history. **Falsified** if unwitnessed presence produces equal downstream gratitude — in which case **the chronicle is decorative**, the loop closes on its own, and the most distinctive half of this architecture is unnecessary. *The authors commit to publishing this either way; it is the prediction whose null would cost the most.*
- **P-L3 (cost, not quantity, drives the return).** Controlling for total hours shared, **intermittent high-cost co-presence** will occasion more expressed gratitude per hour than **steady low-cost co-presence** (e.g. cohabiting pairs). **Falsified** if expression scales with quantity of time rather than with its cost — which would refute §3.1's habituation argument and, with it, the reason the witness position exists.

The pretext prediction (added 2026-08-22, before any of it is built). §4.6 argues that in-person key verification has failed for thirty years on the invitation rather than on the cryptography. That is a behavioral claim and gets an instrument.

- **P-M1 (the gift is the pretext).** Among participants who complete a co-present redemption, the rate of completing an in-person device-to-device verification exchange — offered once, in context, never nagged — will exceed the unaided base rate for authentication ceremonies reported in the secure-messaging literature (14% locating the ceremony unaided; Vaziripour et al., SOUPS 2017) by a wide margin, and will do so **without** the explicit prompting that lifted completion to 90% in the 2018 follow-up. **Falsified** if completion inside a redemption looks like the unaided base rate — which would mean the occasion was never the binding constraint, that prompting was, and that this section's argument is wrong in an interesting way. Publish either way.

Citation note to P-M1 (added 2026-09-26; the prediction's wording and comparator are unchanged): the 14% figure is the 2017 study's first-phase rate, in which participants were instructed about potential threats but not about the ceremony, and it counts attempts that both located *and* completed the ceremony; the 2018 follow-up also simplified the ceremony as well as prompting for it (§4.6).

11 · The Adversarial Surface

The darkest adjacency: paid companionship. A time-currency touching money must answer the pattern it must never become. The answer is structural at four layers: no conversion exists (the wall); no rate can form on any surface (duration-blindness, prompt-suppression, and P-C3 watching); the reconnection core is known-counterparty (one's own mother is not a market); and the stranger-facing layer is a separate mechanism behind an explicit boundary, separately safeguarded, and never doorstep-routing. What remains — off-platform side-arrangements between consenting adults — is outside any system's surfaces; the design's honest claim is that *its* surfaces price nothing and its grammar cannot express the transaction.

Coerced presence. The gift of time must never become its obligation. Guards: availability is expressed as a curated menu (no open-ended asks), mutuality signals are never gates, blocking overrides everything, and the veto is per-event and unpenalized in the moment (only *chronic* refusal surfaces, and only as ledger fact).

Current form. Chronic refusal no longer surfaces at all: no refusal count and no received-versus-delivered gap is displayed (§7.2). The parenthesis above is retained as a disclosed variant.

Attestation fraud. A dyad colluding to fake delivered hours gains: ledger honor with no monetizable payout (delivered-hours confer no currency), and any follower re-thanks their staged content might attract — which is the witness layer's existing integrity problem, met by its existing machinery (human verification, curation, real-impact weighting) rather than a new one. The farming audit conclusion: the redemption layer mints nothing, so the redemption layer cannot be farmed; only the witness layer carries value, and it is already guarded.

Guilt machinery. An expiring currency can torment. The containments: expiry grief is channeled into the lapse-channel's outward gift (guilt metabolized as *dāna*, not as churn or shame); no lapse is ever billed, penalized, or publicly itemized; and the application never prompts the redeemed party toward reciprocation (the no-backward-prompt guard doubles as anti-guilt architecture).

Menu gaming. Attempts to compose extraction inside the co-service class fail the both-hands or outward-benefit conjunction, and anything that passes both while still feeling wrong meets the veto. The residual —

genuinely consensual, genuinely mutual, genuinely outward work that an outside observer would still call lopsided — is a family's own business.

The agent hour-farm (added 2026-08-07). The sharpest adversarial case against a proof-of-sacrifice primitive is two artificial agents in *perfect* co-presence: synchronous, undistracted, attentive without lapse, mutually attested, indefinitely repeatable. **Every observable the gate checks is maximised, and the value is zero** — which is Goodhart's law arriving at presence, and it deserves a direct answer rather than a policy.

The answer is that the observables were never the referent. **The unit is the foregone alternative**, and an agent that can be forked forgoes nothing by attending: the instance that sat with you did not thereby fail to sit with anyone else. **Parallelisability does not make the cost cheap; it removes the scarcity structurally**, so there is no sacrifice for the attestation to be evidence of. §7.1 states the general form.

A second, independent route closes it harder. If inter-agent influence runs only as object-condition and decisive-support edges and never as contiguity, then two agents exchanging turns **share no clock** — and synchronicity was the first load-bearing term in §4.1. On that reading the case is not a counterfeit co-presence; **it is not co-presence at all**, and the gate excludes it definitionally rather than by detection.

The uncomfortable symmetry, stated because it is the honest part. The very property that makes agent presence costless — forkability, and the absence of a single unrepeatable trace — is the property on which this corpus denies such systems the dignity floor it extends to persons. **The line is mortal-and-copyable, and it does the work in both arguments at once.** That is either a deep consistency or a convenient one, and the falsifier of §7.1 applies here too: a single-threaded, unforkable, mortal agent breaks the premise in both places simultaneously.

Mixed pairs are a sink, not a counterfeit. A human who redeems an hour with an agent has spent finite life; the agent has not. Nothing is minted that should not be, and no fraud occurs — but the loop of §3.1 does not turn, because the return leg has no stream to arise in. The correct treatment is therefore not prohibition but accounting: such an event **consumes an hour and produces no witness**, and the design should say so plainly rather than pretend the transaction is symmetric.

Ledger shaming. Earlier drafts answered this with a rule — non-rankable axes, no leaderboards, no percentiles, no feeds of "worst decliners" — which held only for as long as whoever owned the surface chose to hold it. §7.2 replaces the rule with a property: the gap between received and delivered is **not rendered at all**, and expiry does the enforcing. What remains displayable is what a participant has given and what they have received; the difference between them is never computed for display, so there is no quadrant to be shamed into.

12 • Honest Limits

Unbuilt, n = 0. The Chronicle is design-complete and unimplemented; the pilot evidence in this corpus belongs to the money-half. Every mechanism here is strata-dated to the design layer, and the predictions of §10 are the paper's only empirical commitments.

The wall governs surfaces, not side-deals. As with every wall-layer publication in this corpus: architecture can make a rate inexpressible on the system's surfaces and cannot police arrangements struck beside them. Non-fungibility deters the side-deal (an hour of a specific person is a poor commodity); it does not abolish ingenuity. P-C3 is the honest instrument: if a shadow rate forms anyway, the number will say so in public.

Co-presence excludes, at the margin. Synchrony is a real cost — time zones, shift work, connectivity, disability, incarceration all tax it, unevenly. The remote modality and the flanking geometry recover much of the margin (a call redeems; silence side-by-side on a call redeems); what synchrony still excludes, this design accepts as the price of a currency whose entire value is that it cannot be delivered by proxy. A presence-currency with an asynchronous fallback is a messaging app with guilt.

The launch modality is the call, not the room. The diaspora wedge means the flagship redemption at launch is remote — proximity attestation, the richer of the two attestations, arrives with the density the product hopes to create. The design's own success metric would eventually shift the modality mix; at launch, honesty requires saying the market walk in the canonical scene happens through a lens.

Attestation privacy. Proximity attestation is data-minimal by design (a mutual confirmation, not a location trail), but any presence-verification carries surveillance adjacency; the implementation must hold the line that the system learns *that* two people met, never *where they went*.

No guaranteed valence — presence is a high-bandwidth channel, not a good one (added 2026-08-07). This paper has argued throughout that co-presence is where the loop's most valuable conditioning happens. It has not, until now, said the obvious corollary: **a high-bandwidth channel carries whatever is sent through it.** An hour with an estranged parent can condition resentment as readily as reconciliation, and the mechanism has no way to tell which is occurring and no business trying (§4.5's refusal). The design's honest claim is narrower than its rhetoric has sometimes implied: it maximises *conditioning* and guarantees nothing about *valence*. Every surface that promises reconnection should be read against this limit, and the product's copy is the place where the overpromise is most likely to reappear.

The elegance caution. The circuit closes suspiciously well — six steps, two currencies, one wall, every doctrine of this corpus arriving on cue. Internal coherence is the cheapest of all evidence and the most seductive; the authors state, as always, that the elegance earned the specification and only the predictions can earn the product.

13 • Lineage and Close

(Labeled lineage: grounding and translation, never mechanism; §§3–8 stand without it.)

The circuit's joints carry the canon's oldest teaching about gratitude: that it is complete only when enacted — the grateful person the tradition praises is the one who *knows what was done and makes it known*, and the mechanism's every arrow is that completion given a form. The limit case stands where this corpus always places it: the Śāśa Jātaka's hare, who gave the two scarcities whole and at once — the possession-grade and the life-grade of giving converging in a single body — witnessed, the story says, by a sky that kept the image. The Chronicle trades no one toward that summit; it is a foundation practice for householders, denominated in Sunday mornings. But the summit fixes the axis: the economy below it circulates hours and

dollars so that, between two people who had gone dark to each other, the thing the hare gave completely might circulate in installments — one market walk, one mango argument, one answered pin of rejoicing at a time. The currency that cannot be spent alone is, in the end, a machine for making sure that the sentence "we should talk more" sometimes becomes a date.

Terms

Coined names used in this paper and the standard terms an examiner would search for them.

Term used here	Standard technical term
Chronicle ; time-gratitude; pledged hour	time currency; time-banking credit; non-transferable, expiring time-credit issued as thanks between two named persons
co-presence-gated redemption	redemption of a time credit conditioned on a synchronous two-party meeting (in person or live audio/video), attested by both parties
facing / flanking	face-to-face interaction / side-by-side joint attention on a shared task or object
envelope-not-delivery	exclusion of asynchronous media (recorded voice or video messages) from redeeming a credit; such media may only carry an invitation
symmetric-cost redemption (claim 2: symmetric duration)	both parties expend the same span of real time; the quality of attention is not measured
presence-scoring refusal	design constraint: no scoring, rating or inference of engagement or attention quality
recipient-authored, giver-vetoed composition	the credit holder selects the activity from a curated catalogue; the credit issuer holds a per-event right of refusal
both-hands rule / outward-benefit rule	joint performance by both parties / third-party beneficiary — two conditions admitting shared volunteer work as a redemption activity
tense wall ; coupled, never convertible	two currencies (money and time) linked by gratitude flows with no exchange rate or conversion
cross-currency response circuit	a thank-you payment in money for a completed time event, unprompted and independent of the event's duration
witness inversion; sender-default routing	the beneficiary records and shares the event (with both parties' consent); third-party tips route by default to the time-giver
threshold activation; mortal expiry	credits accrue below a threshold, activate on crossing it, and expire one month after activation without recovery
lapse channel	on expiry, an optional anonymous money gift from a common pool to a nearby verified stranger, offered to the issuer
proof-of-sacrifice primitive	attestation that a finite, non-parallelisable span of a person's time was spent with another
Proof of Meeting SM (PoM)	in-person, device-to-device public-key fingerprint exchange (key-signing; Signal safety-number verification)
Proof of HumanitySM	proof of personhood; sybil resistance

Cross-Venue References

Venue	Identifier
Primary canonical	https://thonly.org/research/co-presence-gated-redemption
GitHub	https://github.com/thonly/publications/blob/main/defensive-publications/co-presence-gated-redemption.md
Internet Archive	https://web.archive.org/web/2026*/thonly.org/research/co-presence-gated-redemption

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— End of paper —

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